SIMPLEX METHOD

$$
\begin{array}{llr}
\max & z & \\
\text{st} & z - 3x_1 - 5x_2 = \boxed{0} & (0) \\
& x_1 + x_3 = \boxed{4} & (1) \\
& 2x_2 + x_4 = \boxed{12} & (2) \\
& 3x_1 + 2x_2 + x_5 = \boxed{18} & (3) \\
& x_j \geq 0 \quad j = 1..5 &
\end{array}
$$

- the SIMPLEX METHOD treats the CURRENT VALUE of the OBJECTIVE FUNCTION as another variable
  => at every iteration, the current objective function can be read off from the $\boxed{\text{RHS}}$, together with the values of the basic variables
- At the beginning, we take (x_3, x_4, x_5) as the BASIS
  All other variables are NON-BASIC $(=0)$

  => $x_1 = 0$ $x_3 = 4$ $z = 0$
  $x_2 = 0$ $x_4 = 12$
  $x_5 = 18$

  REMARK: Basic variables do not appear in the objective function!
  (simplex method ensures that this happens at every iteration)
  => if we allow a SINGLE currently NON-BASIC variable to become basic, we can directly see the improvement in z per unit increase of that variable
  => Change in z per unit increase:
  $x_1 \longrightarrow 3$ $(x_1 = 1 \;\; x_2 = 0 \;\; z = 3)$
  $x_2 \longrightarrow 5$ $(x_1 = 0 \;\; x_2 = 1 \;\; z = 5)$

  => x_2 becomes ENTERING variable!
- We allow x_2 to increase until another constraint boundary is reached (i.e., a current basic variable is driven to zero)

HOW? Let's look at the constraints..

We keep $x_1 = 0$ and emphasize on the dependency on x_2

(1) $x_3 = 4$ *
(2) $x_4 = 12 - 2x_2$
(3) $x_5 = 18 - 2x_2$

We keep the other non-basic variable at zero so we can directly see the response of the basic variable to an increase in the entering variable

* not affected by the increase in x_2 because the constraint's boundary it represents is PARALLEL to the direction of increase

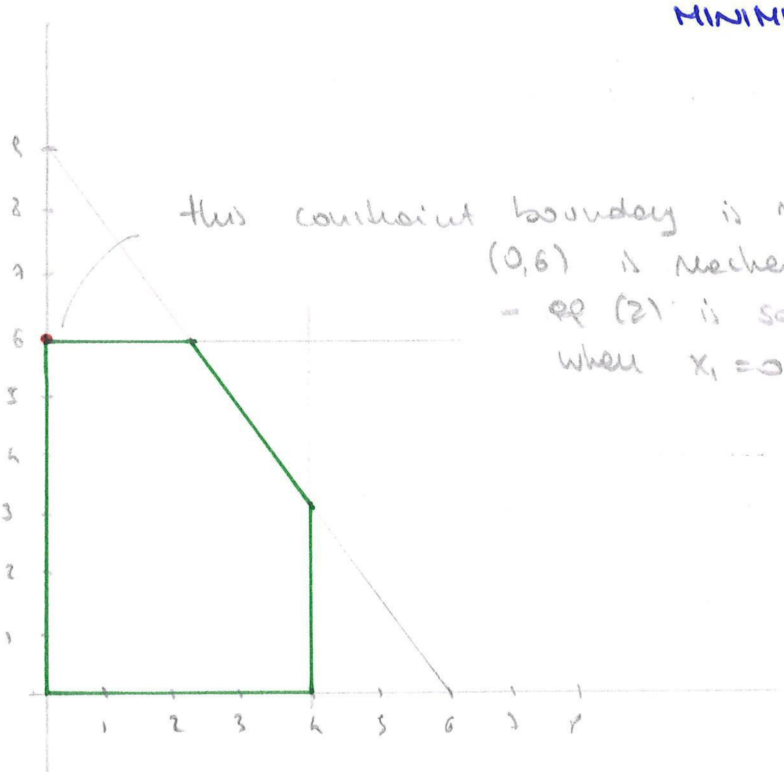
$$
\begin{array}{lllr}
\max & z & & \\
\text{st} & z - 3x_1 - 5x_2 & = 0 & (0) \\
 & x_1 + x_3 & = 4 & (1) \\
 & 2x_2 + x_4 & = 12 & (2) \\
 & 3x_1 + 2x_2 + x_5 & = 18 & (3)
\end{array}
$$

(1) $x_3 = 4$

(2) $x_4 = 12 - 2x_2 \rightarrow 12/2 = 6$ $\Rightarrow$ the first basic variable to be driven to 0 is x_4!

(3) $x_5 = 18 - 2x_2 \rightarrow 18/2 = 9$

MINIMUM RATIO TEST



$\Rightarrow$ x_2 ENTERS

x_4 LEAVES

we have to update the TABLEAU (i.e., the equations) to reflect the new BFS

$\Rightarrow$ GAUSSIAN ELIMINATION (i.e., row operations)

$$
\begin{array}{c}
\hspace{3cm} A \hspace{4cm} b \\
\left[\begin{array}{cccccc}
1 & -3 & -5 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 2 & 0 & 1 & 0 \\
0 & 3 & 2 & 0 & 0 & 1
\end{array}\right]
\left[\begin{array}{c} 0 \\ 4 \\ 12 \\ 18 \end{array}\right] \\
\begin{array}{cccccc} z & x_1 & x_2 & x_3 & x_4 & x_5 \end{array}
\end{array}
$$

$R_0 \rightarrow R_0 + \frac{5}{2} R_2$ (Because x_2 should not be in the obj fun)

$R_2 \rightarrow R_2/2$ (Because x_2 enters the basis)

$R_3 \rightarrow R_3 - R_2$ (x_2 should be the only 1 in its column)

$$
\left[\begin{array}{cccccc}
1 & -3 & 0 & 0 & 5/2 & 0 \\
0 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 1/2 & 0 \\
0 & 3 & 0 & 0 & -1 & 1
\end{array}\right]
\left[\begin{array}{c} 30 \\ 4 \\ 6 \\ 6 \end{array}\right]
$$

$$
\begin{array}{cccccc} & & \uparrow & \uparrow & & \uparrow \end{array}
$$

NEW BASIS: $\{x_2, x_3, x_5\}$, hence, $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$

NEW VALUE for OBJ FUN: $z = 30$

- Are we finished?

  NO!

- non-basic variable x_1 still has potential to improve the objective function
  (the fact that the coeff of x_4 is non-negative means that x_4 cannot improve the objective function)
  - if you move it to the RHS, the RHS will decrease

$\Rightarrow$ x_1 ENTERING VARIABLE $(x_4 = 0)$

which variable will leave?

$$
\begin{array}{llllllll}
\max & z & & & & & & \\
\text{st} & z & -3x_1 & & & +\frac{5}{2}x_4 & & = 30 & (0) \\
& & x_1 & & +x_3 & & & = 4 & (1) \\
& & & x_2 & & +\frac{1}{2}x_4 & & = 6 & (2) \\
& & 3x_1 & & & -x_4 & +x_5 & = 6 & (3)
\end{array}
$$

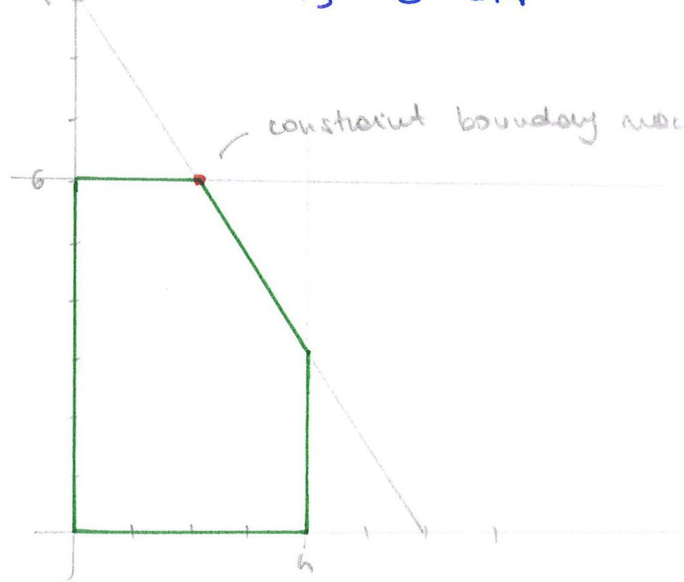
(1) $x_3 = 4 - x_1$ $\rightarrow$ $4/1 = 4$

(2) x_2 un-affected by an increase in x_1

(3) $x_5 = 6 - 3x_1$ $\rightarrow$ $6/3 = 2$

$\uparrow$

x_5 leaving basic variable



$\Rightarrow$ GAUSSIAN ELIMINATION to have the 0.1 pattern in the column x_1 and to remove it from the objective function

$$
\begin{array}{cc}
A & b \\
\begin{bmatrix} 1 & -3 & 0 & 0 & 5/2 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 \\ 0 & 3 & 0 & 0 & -1 & 1 \end{bmatrix} & \begin{bmatrix} 30 \\ 4 \\ 6 \\ 6 \end{bmatrix}
\end{array}
$$

$R_0 \rightarrow R_0 + R_4$

$R_1 \rightarrow R_1 - \frac{1}{3}R_4$

$R_3 \rightarrow R_3/3$

$$
\begin{bmatrix} 1 & 0 & 0 & 0 & 3/2 & 1 \\ 0 & 0 & 0 & 1 & 1/3 & -1/3 \\ 0 & 0 & 1 & 0 & 1/2 & 0 \\ 0 & 1 & 0 & 0 & -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 36 \\ 2 \\ 6 \\ 2 \end{bmatrix}
$$

NEW BASIS: $\{x_1, x_2, x_3\}$

$\Rightarrow (x_1, x_2, x_3, x_4, x_5) = (2, 6, 2, 0, 0)$

NEW VALUE for OBJ FUN: $z = 36$

ARE WE DONE?

yey !! :)

Neither of the two non basic variables can improve the objective function

=> we have reached optimality!

SIMPLEX METHOD

1. Can a currently non-basic variable help to improve the objective function?
2. NO. DONE!
3. YES. Find the non-basic variable with the most potential to improve the objective function

   => ENTERING BASIC VARIABLE
4. Use MINIMUM RATIO TEST to identify the LEAVING BASIC VARIABLE
5. Update the system for the new BFS using GAUSSIAN ELIMINATION
6. GO to 1.

4.5/6/8
5.4/8

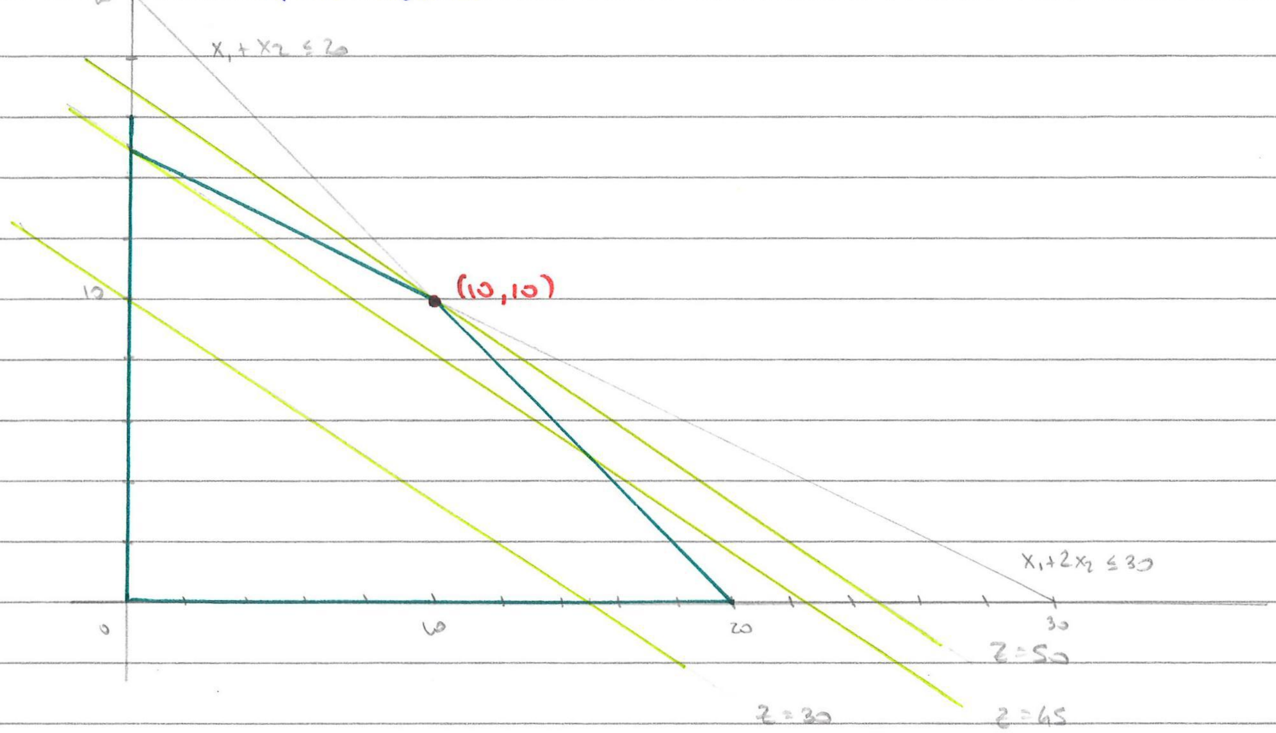
Ex 4.4-S4

$\max \; Z = 2x_1 + 3x_2$ $\qquad 2x_1 + 3x_2 = 30$

st $\qquad x_1 + 2x_2 \leq 30$ $\qquad 20 + 30 = 50$

$\qquad x_1 + x_2 \leq 20$

$\qquad x_1 \geq 0 \quad x_2 \geq 0$



AUGMENTED FORM: $\max \; Z = 2x_1 + 3x_2$ (0)

st $\qquad x_1 + 2x_2 + x_3 = 30$ (1)

$\qquad x_1 + x_2 + x_4 = 20$ (2)

$\qquad x_j \geq 0 \quad j = 1..4$

ALGEBRAIC SIMPLEX:

- STARTING POINT: ORIGIN $(0, 0, 30, 20)$
- ENTERING BASIC VARIABLE: max increase in obj function $\Rightarrow x_2$
- LEAVING BASIC VARIABLE: min ratio test $\quad 30/2 = 15 \Rightarrow x_3$
  $\quad 20/1 = 20$
- CURRENT POINT: $(0, 15, 0, 5)$
  $(0, 15)$ CPF SOL

(0) $z - 2x_1 - 3x_2 = 0$

(1) $x_1 + 2x_2 + x_3 = 30$

(2) $x_1 + x_2 + x_4 = 20$

becomes

(0) $z - \frac{1}{2}x_1 + \frac{3}{2}x_3 = 45$ (x_2 have coeff 0) Eq(0)-3Eq(1)

(1) $\frac{1}{2}x_1 + x_2 + \frac{1}{2}x_3 = 15$ (x_2 have coeff 1) 1/2 Eq(1)

(2) $\frac{1}{2}x_1 - \frac{1}{2}x_3 + x_4 = 5$ (x_2 have coeff 0) Eq(2) - Eq(1)

OPTIMALITY TEST: NO! x_1 can still improve the obj

ENTERING BASIC VARIABLE: x_1

LEAVING BASIC VARIABLE: $15/\frac{1}{2} = 30$

$5/\frac{1}{2} = 10 \Rightarrow x_4$

CURRENT POINT: (10,10,0,0)

(0) $z + x_3 + x_4 = 50$

(1) $x_2 + x_3 - x_4 = 10$

(2) $x_1 - x_3 + 2x_4 = 10$ (x_1 have coeff 1)

OPTIMALITY TEST: YES!

EX 4.4.5

max $z = 5x_1 + 8x_2 + 7x_3$

$x_1 + 3x_2 + 2x_3 \leq 10$

$3x_1 + 4x_2 + 2x_3 \leq 12$

$2x_1 + x_2 + 2x_3 \leq 8$

$x_i \geq 0 \quad i = 1,2,3$

AUGMENTED FORM max z

$$
\begin{aligned}
z - 5x_1 - 8x_2 - 7x_3 \qquad\qquad\qquad &= 0 \quad (0)\\
x_1 + 3x_2 + 2x_3 + x_4 \qquad\qquad &= 10 \quad (1)\\
3x_1 + \boxed{4x_2} + 2x_3 \qquad + x_5 \qquad &= 12 \quad (2)\\
2x_1 + x_2 + 2x_3 \qquad\qquad + x_6 &= 8 \quad (3)
\end{aligned}
$$

$x_i \geq 0 \quad i = 1 \dots 6$

STARTING POINT: ORIGIN $(0,0,0,10,12,8)$

BASIS $\{x_4, x_5, x_6\}$

ENTERING BASIC VARIABLE : x_2 (max increase in cost - most negative coeff in Eq (0))

LEAVING BASIC VARIABLE : knowing that $x_1 = x_3 = 0$

(1) $x_4 = 10 - 3x_2$ $\quad 10/3 \sim 1,3$

(2) $x_5 = 12 - 4x_2$ $\quad 12/4 = 3$

(3) $x_6 = 8 - x_2$ $\quad 8/1 = 8$

MINIMUM RATIO TEST : x_5

UPDATE THE TABLEAU to HAVE $\boxed{1}$ in the GREEN SPOT and 0 in the REST of the COLUMN

USING GAUSSIAN ELIMINATION : Eq (2) $\longrightarrow$ Eq (2)/4

Eq (0) $\longrightarrow$ Eq (0) + 8 Eq (2)

Eq (1) $\longrightarrow$ Eq (1) − 3 Eq (2)

Eq (3) $\longrightarrow$ Eq (3) − Eq (2)

$$
\begin{array}{llllllll}
(0) & z - \frac{7}{4}x_1 & & - \frac{5}{2}x_3 & & + \frac{9}{4}x_5 & & = 27 \\
(1) & \quad - \frac{5}{4}x_1 & & \boxed{+ \frac{1}{2}x_3} & + x_4 & - \frac{3}{4}x_5 & & = 1 \\
(2) & \quad \frac{3}{4}x_1 & + x_2 & + \frac{1}{2}x_3 & & + \frac{1}{4}x_5 & & = 3 \\
(3) & \quad \frac{5}{4}x_1 & & + \frac{3}{2}x_3 & & - \frac{1}{4}x_5 & + x_6 & = 5
\end{array}
$$

CURRENT POINT: $x_4 = 1$ $\quad x_2 = 3$ $\quad x_6 = 5$ (BASIS)

$x_1 = 0$ $\quad x_3 = 0$ $\quad x_5 = 0$

$(0, 3, 0, 1, 0, 5)$

OPTIMAL? NO! still negative term in Eq(0)

ENTERING BASIC VARIABLE: x_3 (max increase in cost - most negative coeff in Eq(0))

LEAVING BASIC VARIABLE: $x_1 = x_5 = 0$

$$
\begin{array}{ll}
x_4 = 1 - \frac{1}{2}x_3 & 1/\frac{1}{2} = 2 \\
x_2 = 3 - \frac{1}{2}x_3 & 3/\frac{1}{2} = 6 \\
x_6 = 5 - \frac{3}{2}x_3 & 5/\frac{3}{2} = 10/3 \sim 3
\end{array}
$$

$\Rightarrow x_4$

UPDATE TABLEAU: Eq(1) $\rightarrow$ Eq(1) $\cdot\, 2$

Eq(0) $\rightarrow$ Eq(0) $+ \frac{5}{2}$ Eq(1)

Eq(2) $\rightarrow$ Eq(2) $- \frac{3}{2}$ Eq(1)

Eq(3) $\rightarrow$ Eq(3) $- \frac{3}{2}$ Eq(1)

$$
\begin{array}{llllllll}
(0) & z - \frac{9}{2}x_1 & & & + 5x_4 & - \frac{3}{2}x_5 & & = 32 \\
(1) & \quad - \frac{5}{2}x_1 & & + x_3 & + 2x_4 & - \frac{3}{2}x_5 & & = 2 \\
(2) & \quad 2x_1 & + x_2 & & - x_4 & + x_5 & & = 2 \\
(3) & \quad \boxed{5x_1} & & & - 3x_4 & + 2x_5 & + x_6 & = 2
\end{array}
$$

CURRENT POINT: $x_3 = 2$ $\quad x_2 = 2$ $\quad x_6 = 2$ (BASIS)

$x_1 = 0$ $\quad x_4 = 0$ $\quad x_5 = 0$

$(0, 2, 2, 0, 0, 2)$

OPTIMAL? NO! objective function can still increase

ENTERING: x_1 (max increase in cost - most negative coeff)

LEAVING: $x_4 = x_5 = 0$

$x_3 = 2 + \frac{5}{2} x_1$ IGNORE THIS! WHY?

$x_2 = 2 - 2x_1$ $2/2 = 1$

$x_6 = 2 - 5x_1$ $2/5 = 0.4$

$\Rightarrow$ x_6

UPDATE TABLEAU: $Eq(3) \rightarrow Eq(3)/5$

$Eq(0) \rightarrow Eq(0) + \frac{18}{4} Eq(3)$

$Eq(1) \rightarrow Eq(1) + \frac{5}{2} Eq(3)$

$Eq(2) \rightarrow Eq(2) - 2\,Eq(3)$

$$
\begin{array}{llllll}
(0) & z & & & + \frac{23}{10} x_4 + \frac{3}{10} x_5 + \frac{9}{10} x_6 & = \frac{168}{5} \\
(1) & & & + x_3 & + \frac{1}{2} x_4 + \frac{1}{2} x_5 + \frac{1}{2} x_6 & = 3 \\
(2) & & + x_2 & & + \frac{1}{5} x_4 + \frac{1}{5} x_5 - \frac{2}{5} x_6 & = \frac{6}{5} \\
(3) & x_1 & & & - \frac{3}{5} x_4 + \frac{2}{5} x_5 + \frac{1}{5} x_6 & = \frac{2}{5}
\end{array}
$$

CURRENT POINT: $x_1 = \frac{2}{5}$ $x_2 = \frac{6}{5}$ $x_3 = 3$

$x_4 = x_5 = x_6 = 0$

$\left(\frac{2}{5}, \frac{6}{5}, 3, 0, 0, 0\right)$

OPTIMAL? YES !! :) $z = \frac{168}{3} \sim 56$

EX 4.4.6

max $Z = 3x_1 + 5x_2 + 6x_3$

$$\begin{aligned}
2x_1 + x_2 + x_3 &\le 4 \\
x_1 + 2x_2 + x_3 &\le 4 \\
x_1 + x_2 + 2x_3 &\le 4 \\
x_1 + x_2 + x_3 &\le 3 \\
x_j &\ge 0
\end{aligned}$$

AUGMENTED FORM:

$$\begin{array}{llr}
Z - 3x_1 - 5x_2 - 6x_3 & = 0 & (0) \\
2x_1 + x_2 + x_3 + x_4 & = 4 & (1) \\
x_1 + 2x_2 + x_3 + x_5 & = 4 & (2) \\
x_1 + x_2 + \boxed{2x_3} + x_6 & = 4 & (3) \\
x_1 + x_2 + x_3 + x_7 & = 3 & (4)
\end{array}$$

STARTING POINT: ORIGIN $(0, 0, 0, 4, 4, 4, 3)$

OPTIMAL ? NO!

ENTERING BASIC VARIABLE: x_3

LEAVING BASIC VARIABLE:

$x_4 = 4 - x_3$ $\quad 4/1 = 4$

$x_5 = 4 - x_3$ $\quad 4/1 = 4$

$x_6 = 4 - 2x_3$ $\quad 4/2 = 2$

$x_7 = 3 - x_3$ $\quad 3/1 = 3$

$\Rightarrow x_6$

UPDATE TABLEAU:

$Eq(3) \to Eq(3)/2$

$Eq(0) \to Eq(0) + 6\,Eq(3)$

$Eq(1) \to Eq(1) - Eq(3)$

$Eq(2) \to Eq(2) - Eq(3)$

$Eq(4) \to Eq(4) - Eq(3)$

$$
\begin{array}{c}
\begin{array}{cccccccc|c}
z & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 & \text{RHS}
\end{array} \\
\left[\begin{array}{cccccccc|c}
1 & 0 & -2 & 0 & 0 & 0 & 3 & 0 & 12 \\
0 & 3/2 & 1/2 & 0 & 1 & 0 & -1/2 & 0 & 2 \\
0 & 1/2 & \boxed{3/2} & 0 & 0 & 1 & -1/2 & 0 & 2 \\
0 & 1/2 & 1/2 & 1 & 0 & 0 & 1/2 & 0 & 2 \\
0 & 1/2 & 1/2 & 0 & 0 & 0 & -1/2 & 1 & 1
\end{array}\right]
\begin{array}{c}
(0) \\ (1) \\ (2) \\ (3) \\ (4)
\end{array}
\end{array}
$$

CURRENT POINT : $x_4 = 2 \quad x_5 = 2 \quad x_3 = 2 \quad x_7 = 1$

$x_1 = x_2 = x_6 = 0$

$(0, 0, 2, 2, 2, 0, 1)$

OPTIMAL ? NO !

ENTERING : x_2

LEAVING : $x_4 = 2 - \frac{1}{2} x_2 \qquad 2 / \frac{1}{2} = 4$

$x_5 = 2 - \frac{3}{2} x_2 \qquad 2 / \frac{3}{2} = 4/3 \sim 1{,}3$

$x_3 = 2 - \frac{1}{2} x_2 \qquad 2 / \frac{1}{2} = 4$

$x_7 = 1 - \frac{1}{2} x_2 \qquad 1 / \frac{1}{2} = 2$

$\Rightarrow x_5$

UPDATE TABLEAU: $Eq(2) \to Eq(2) \cdot \frac{2}{3}$

$Eq(0) \to Eq(0) + 2\, Eq(2)$

$Eq(1) \to Eq(1) - \frac{1}{2} Eq(2)$

$Eq(3) \to Eq(3) - \frac{1}{2} Eq(2)$

$Eq(4) \to Eq(4) - \frac{1}{2} Eq(2)$

$$
\begin{array}{c}
\begin{array}{cccccccc|c}
z & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 &
\end{array} \\
\left[\begin{array}{cccccccc|c}
1 & 2/3 & 0 & 0 & 0 & 4/3 & 7/3 & 0 & 44/3 \\
0 & 4/3 & 0 & 0 & 1 & -1/3 & -1/3 & 0 & 4/3 \\
0 & 1/3 & 1 & 0 & 0 & 2/3 & -1/3 & 0 & 4/3 \\
0 & 1/3 & 0 & 1 & 0 & -1/3 & 2/3 & 0 & 4/3 \\
0 & 1/3 & 0 & 0 & 0 & -1/3 & -1/3 & 1 & 1/3
\end{array}\right]
\end{array}
$$

CURRENT POINT : $x_4 = 4/3 \quad x_2 = 4/3 \quad x_3 = 4/3 \quad x_7 = 4/3$

$x_1 = 0 \quad x_5 = 0 \quad x_6 = 0$

$(0, \frac{4}{3}, \frac{4}{3}, \frac{4}{3}, 0, 0, \frac{4}{3})$

OPTIMAL? YES !!

Ex 4.5.4 (unbounded)

max $Z = 5x_1 + x_2 + 3x_3 + 4x_4$

$$x_1 - 2x_2 + 4x_3 + 3x_4 \le 20$$
$$-4x_1 + 6x_2 + 5x_3 - 4x_4 \le 40$$
$$2x_1 - 3x_2 + 3x_3 + 8x_4 \le 50$$
$$x_j \ge 0$$

AUGMENTED FORM :

$$\begin{array}{llr}
Z - 5x_1 - x_2 - 3x_3 - 4x_4 & = 0 & (0) \\
\boxed{x_1} - 2x_2 + 4x_3 + 3x_4 + x_5 & = 20 & (1) \\
-4x_1 + 6x_2 + 5x_3 - 4x_4 + x_6 & = 40 & (2) \\
2x_1 - 3x_2 + 3x_3 + 8x_4 + x_7 & = 50 & (3)
\end{array}$$

START: (0, 0, 0, 0, 20, 40, 50)

ENTERING BASIC VARIABLE: x_1 (max increase)

LEAVING BASIC VARIABLE: $x_5 = 20 - x_1$ $20/1 = 20$

$x_6 = 40 + 4x_1$ IGNORE

$x_7 = 50 - 2x_1$ $50/2 = 25$

$\Rightarrow x_5$

$$\begin{array}{c}
(0) \\ (1) \\ (2) \\ (3)
\end{array}
\left[\begin{array}{cccccccc|c}
1 & 0 & -11 & 17 & 11 & 5 & 0 & 0 & 100 \\
0 & 1 & -2 & 4 & 3 & 1 & 0 & 0 & 20 \\
0 & 0 & -2 & 21 & 8 & 4 & 1 & 0 & 120 \\
0 & 0 & \boxed{1} & -5 & 2 & -2 & 0 & 1 & 10
\end{array}\right]
\begin{array}{l}
R_0 + 5R_1 \\ \\ R_2 + 4R_1 \\ R_3 - 2R_1
\end{array}$$

CURRENT POINT: (20, 0, 0, 0, 0, 120, 10)

OPTIMAL ? NO!

ENTERING BASIC VARIABLE: x_2

LEAVING BASIC VARIABLE: $x_1 = 20 + 2x_2$ IGNORE

$x_6 = 120 + 2x_2$ IGNORE

$x_7 = 10 - x_1$

$\Rightarrow x_7$

$$\left[\begin{array}{cccccccc|c} 1 & 0 & 0 & -38 & 33 & -17 & 0 & 11 & 210 \\ 0 & 1 & 0 & -6 & 7 & -3 & 0 & 2 & 40 \\ 0 & 0 & 0 & \boxed{11} & 12 & 0 & 1 & 2 & 140 \\ 0 & 0 & 1 & -5 & 2 & -2 & 0 & 1 & 10 \end{array}\right] \begin{array}{l} R_0 + 11R_3 \\ R_1 + 2R_3 \\ R_2 + 2R_3 \\ \end{array}$$

CURRENT POINT: $(40, 10, 0, 0, 0, 140, 0)$

OPTIMAL ? NO!

ENTERING BASIC VARIABLE: $\underline{x_3}$

LEAVING BASIC VARIABLE: $x_1 = 40 + 6x_3$ IGNORE

$x_6 = 140 - 11x_3$

$x_2 = 10 + 5x_3$ IGNORE

$\Rightarrow \underline{x_6}$

$$\left[\begin{array}{cccccccc|c} 1 & 0 & 0 & 0 & \frac{819}{11} & -17 & \frac{38}{11} & \frac{197}{11} & \frac{7630}{11} \\ 0 & 1 & 0 & 0 & \frac{149}{11} & -3 & \frac{6}{11} & \frac{34}{11} & \frac{1280}{11} \\ 0 & 0 & 0 & 1 & \frac{12}{11} & 0 & \frac{1}{11} & \frac{2}{11} & \frac{140}{11} \\ 0 & 0 & 1 & 0 & \frac{82}{11} & -2 & \frac{5}{11} & \frac{21}{11} & \frac{810}{11} \end{array}\right] \begin{array}{l} R_0 + 38R_2 \\ R_1 + 6R_2 \\ R_2/11 \\ R_3 + 5R_2 \end{array}$$

WE CAN SEE FROM THE LAST TWO ITERATIONS THAT, BECAUSE ALL CONSTRAINT'S COEFFICIENTS of x_5 ARE NON-POSITIVE (≤ 0), x_5 CAN BE INCREASED WITHOUT FORCING ANY BASIC VARIABLE TO ZERO.

E.G. LAST ITERATION: $x_1 = \frac{1280}{11} + 3x_5$

$x_2 = \frac{810}{11} + 2x_5$

$x_3 = \frac{140}{11}$ (not even affected)

$\Rightarrow$ WHATEVER VALUE of x_5 (AS LONG AS NON-NEGATIVE) IS FEASIBLE AND

$z = \frac{7630}{11} + 17x_5$

EX 4.5.8 (find all optimal sol)

$$\max \quad Z = 50x_1 + 25x_2 + 20x_3 + 40x_4$$

$$\text{st} \quad 2x_1 + x_2 \leq 30$$

$$x_3 + 2x_4 \leq 20$$

$$x_j \geq 0 \quad j = 1-4$$

AUGMENTED FORM: $Z - 50x_1 - 25x_2 - 20x_3 - 40x_4 = 0$

$2x_1 + x_2 + x_5 = 30$

$x_3 + 2x_4 + x_6 = 20$

$x_j \geq 0 \quad j = 1-6$

CURRENT POINT: (0,0,0,0, 30, 60)

ENTERING BV: x_1

LEAVING BV: $x_5 = 30 - 2x_1$

$x_6 = 20 - x_3 - 2x_4$ not affected

$\Rightarrow x_5$

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & -20 & -40 & 25 & 0 & 750 \\ 0 & 1 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 15 \\ 0 & 0 & 0 & 1 & 2 & 0 & 1 & 20 \end{array}\right] \quad R_0 + 50R_1$$

CURRENT POINT: (15, 0, 0, 0, 0, 20)

ENTERING BV : x_4

LEAVING : x_6 (others not affected)

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & 25 & 20 & 1150 \\ 0 & 1 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 15 \\ 0 & 0 & 0 & \frac{1}{2} & 1 & 0 & \frac{1}{2} & 10 \end{array}\right] \quad R_0 + 40R_2$$

CURRENT POINT: (15, 0, 0, 10, 0, 0)

OPTIMAL ? YES

BUT : x_2 and x_3 have 0 coeff in row(0)

$\Rightarrow$ we can pick them as ENTERING BV to get other optimal solution

ENTERING: X_2

LEAVING: X_1 (others not affected)

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & 25 & 20 & 1150 \\ 0 & 2 & 1 & 0 & 0 & 1 & 0 & 30 \\ 0 & 0 & 0 & \frac{1}{2} & 1 & 0 & \frac{1}{2} & 10 \end{array}\right]$$

CURRENT POINT: $(0, 30, 0, 10, 0, 0)$

OPTIMAL? YES!

ENTERING: X_3

LEAVING: X_4

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & 25 & 20 & 1150 \\ 0 & 2 & 1 & 0 & 0 & 1 & 0 & 30 \\ 0 & 0 & 0 & 1 & 2 & 0 & 1 & 20 \end{array}\right]$$

CURRENT POINT: $(0, 30, 20, 0, 0, 0)$

OPTIMAL? YES!

ENTERING: X_1

LEAVING: X_2

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & 25 & 20 & 1150 \\ 0 & 1 & \frac{1}{2} & 0 & 0 & 1/2 & 0 & 15 \\ 0 & 0 & 0 & 1 & 2 & 0 & 1 & 20 \end{array}\right]$$

CURRENT POINT: $(15, 0, 20, 0, 0, 0)$